

RPG (With SFML GUI)

Group Count: 3–4

Explanation:

You are tired of "pay-to-win" mechanics and uninspired AAA sequels. As a developer who values pure gameplay, you've decided to build your own RPG engine from scratch. Your goal: defeat the mechanical menace, Dr. Vroomstein, and his waves of polymorphic minions. Like any true craftsman, you refuse to take shortcuts—this engine must be modular, efficient, and entirely your own.

Structure:

Avatar (abstract class):

This is the foundation for all the characters within the game whether it follows the good side or the dark side. All characters have a unique name, a base speed and max health. You would also need to keep track of current health and position on the screen (don't let the characters escape the computer screen no one wants Ra-one to come to reality).

Move/Moveset:

All moves have unique animations and all moves deal some damage and have a cooldown period. It could be melee (limited range but usually higher damage), ranged (little less damage but longer range) or special move (special moves have the most damage but also come with a long recharge period).

Hero:

Your game must have at least three options for the hero all with their unique sprites, health, base speed and moveset. Each hero must have their unique set of moves. (1 melee, 1 range, 1 special). A move must have a name and all the moves in the game have some unique attribute. During gameplay, the S, D, F keys can be used to call the melee, range or special move of the player. Come up with a smart and modular damage dealing and pointing system to keep the game engaging and enjoyable.

Villain:

This is the face of evil that our hero must destroy to save the world from its wrath. A high-HP character that has a randomized attack pattern. He may also call minions to his help who move in constant unique formations to hit and deal damage to the hero.

Minions:

Depending on the theme, these could be obstacles, minions, objects thrown. They deal damage on clash with our hero.

Theme:

The theme of the game (character sprites, attack animations, overall vibe) can be customized. You may go for a space-theme battle, alien invasion, fairies, dragons or any characters from your favourite sci-fi /

fantasy season or movie. You are also encouraged to find ways to keep this game interesting such as use levels, difficulty modes, random powerups (like shield or higher attacks), constant messages based on what's happening on the screen, structured minion formations and unique/structured movements of those characters (maybe high time to recall the Motions chapter from your Physics book).